

Humanizing· Logistics

Lindsay Daphne Macuja
ARCHIP – AD2
Second Semester
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ARCHIP

ARCHITECTURAL
INSTITUTE
IN PRAGUE

ABOUT ME



Lindsay Daphne Macuja

Hi! I'm Lindsay Daphne Macuja and I'm from the Philippines. I'm a yellow person who loves to learn new things, making, designing, modeling and many things when it comes to creativity and expressing an art and my motto in life is not to be scared with colors, colors is life.

ABOUT THE STUDIO

This semester, the studio reimagined logistics as urban assets rather than isolated infrastructures. Working with Ramboll, Henning Larsen, Panattoni, the Urban Land Institute, and Coding ARCH, we developed proposals grounded in real industry conditions.

My work focused on evidence-based design, using performance data and parametric evaluation to link architectural decisions with outcomes such as energy use, biodiversity, stormwater performance, embodied carbon, and cost.

In the final weeks, my focus centered on shaping a human-scale logistics campus through modularity, KPI development, and LCA-driven material improvements, integrating logistics with public edges and green corridors. The strategy relied on quantitative reasoning and interdisciplinary collaboration rather than visuals alone.

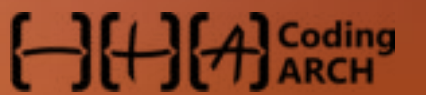


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I.

THE SITE

Site Analysis

Users & Businesses

The Baseline

SITE ANALYSIS

📍 Strojírenská 175/25, 155 21 Zličín, Prague, Czech Republic

The site at Strojírenská 175/25, Zličín is located within a mixed urban–industrial district where a variety of programs coexist in close proximity. The surrounding area includes warehouses and storage facilities, business parks, offices, residential buildings, schools, and local sports facilities. This combination creates a diverse but fragmented urban fabric.

The site is well connected through multiple mobility systems, including a railway line, bus stops, train access, and an established network of walking paths. Parking areas and service roads support the existing logistics and business operations.

Despite this connectivity, the district experiences several challenges typical of logistics zones, such as noise, traffic congestion, heavy vehicle movement, and close adjacency to residential areas. These conditions shape the everyday experience of the site and define the baseline context for any future development.

- warehouse, storages, businesses

● residence

● offices

● school

● business parks

● residential social services

● local sports facilities

● storage, distribution
- walking path

P parking

🚆 railway

■ bus stop

✕ train stop



USERS

The user groups represent the everyday activities that shape the district's character. They include logistics workers, car service technicians, construction and machinery operators, home-improvement and shading specialists, natural product producers, water-treatment providers, heating and cooling technicians, pet-gear makers, textile and branding services, and travel and mobility customers. These users reflect the human side of logistics people who work, repair, produce, and move through the site daily.



Logistics, Storage & Warehousing

Car Service

Construction & Machinery

Home & Shading

Natural Products

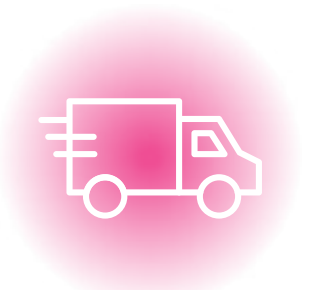
Water Treatment

Heating & Cooling

Pet & Dog Gear

Textile & Branding

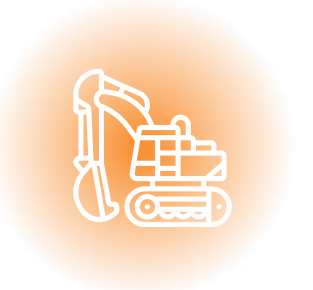
Travel & Mobility



CarMax s.r.o.
BLK servis, spol. s r.o.
HANES, s.r.o.
ROCKTECHNIK, s.r.o.
F.K.M. Praha



Autoservis Kolář
Dědek Autoservis
Autoservis BR
Hybrid Servis



STAS, spol. s r.o.
ROCKTECHNIK, s.r.o.
F.K.M. Praha



ŽALUZIE JEŘÁBEK
MKB Praha, s.r.o.
Kulina Showroom



Přírodní kosmetika
Šebková



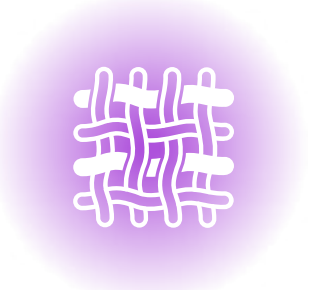
WATERA Czech
UNIVERZÁLNÍ ČISTÁ VODA



HEAT (AC units)
HEAT, s.r.o. (heaters)
MULTI KLIMA, s.r.o.



SOFA Dog Wearr



L-SHOP-TEAM Česko



EUROPE CAMPERS

BUSINESSES

Each user group is supported by a network of local businesses, from logistics and warehousing companies to car repair shops, construction suppliers, home-shading showrooms, natural cosmetics producers, water-treatment firms, HVAC specialists, pet-gear manufacturers, textile distributors, and mobility rental companies. Together, they form a dense economic cluster that depends on efficient access, flexible space, and shared infrastructure.

THE BASELINE

The current baseline for urban logistics presents a fragmented and inefficient model that actively degrades the surrounding urban fabric. Sited inappropriately close to sensitive residential zones and schools, these conventional facilities subject the community to continuous noise pollution from engines and daily loading operations. A severe lack of adequate on-site parking forces heavy trucks onto public streets, while restricted, single-point access creates chronic traffic bottlenecks and unsafe conditions for pedestrians and cyclists. Furthermore, these opaque, "big box" structures completely lack green infrastructure and community interface, resulting in high carbon emissions, excessive heat and dust, and an intimidating, unwelcoming presence within the neighborhood.

Big Box



The existing site is dominated by a massive, monolithic industrial structure encompassing a vast footprint of 13,028.76 m². Architecturally, the building is defined by a repetitive sawtooth roof profile a traditional industrial feature originally intended for factory daylighting and sprawling, opaque facades. Operating as a conventional "big box" warehouse, its sheer scale and impermeable walls create a stark physical barrier within the neighborhood, offering zero human-scale interface or meaningful integration with the surrounding urban grain.

Entrance & Exit



Offices



View of Residence



Trucks



II.

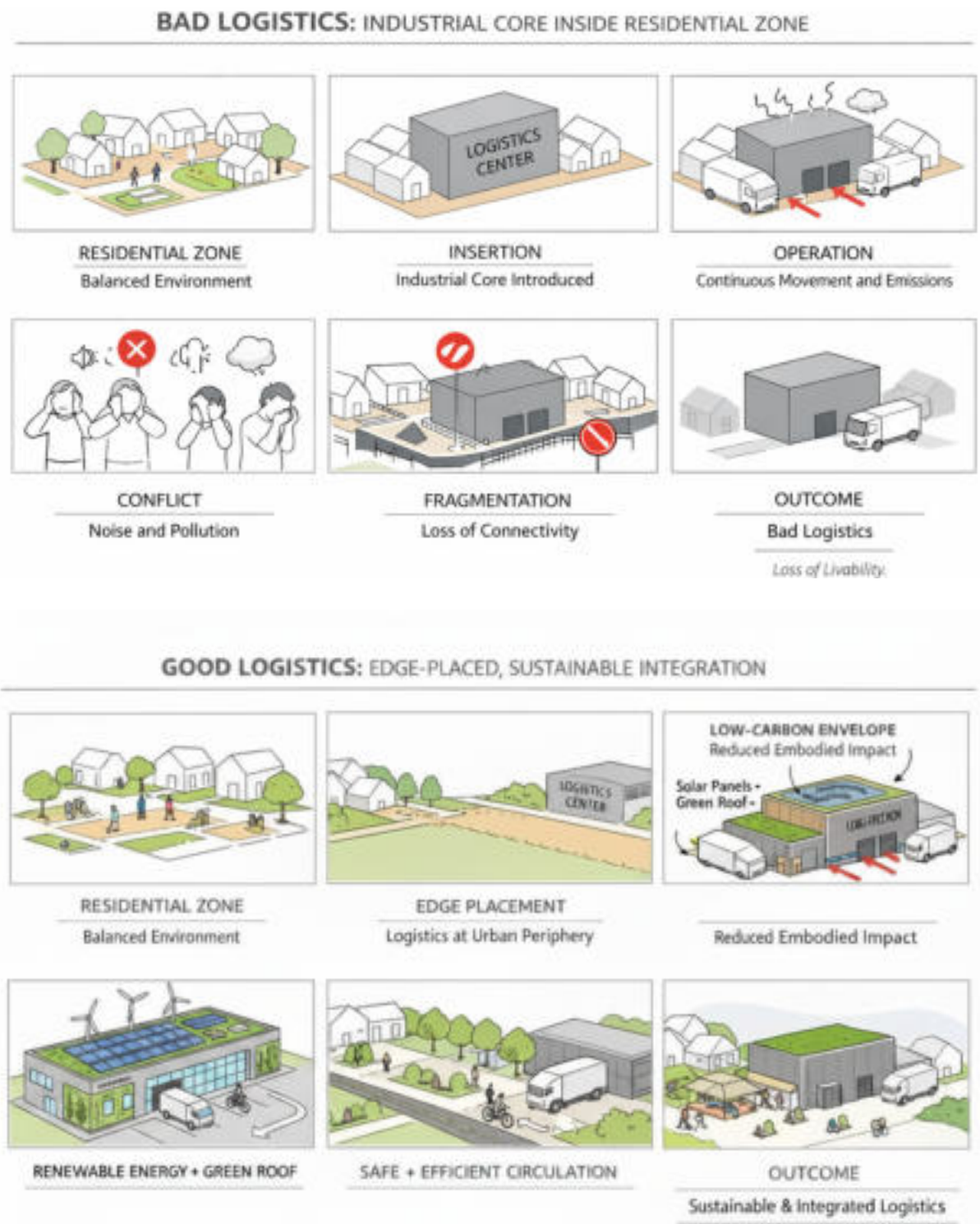
CONCEPT

Case Study

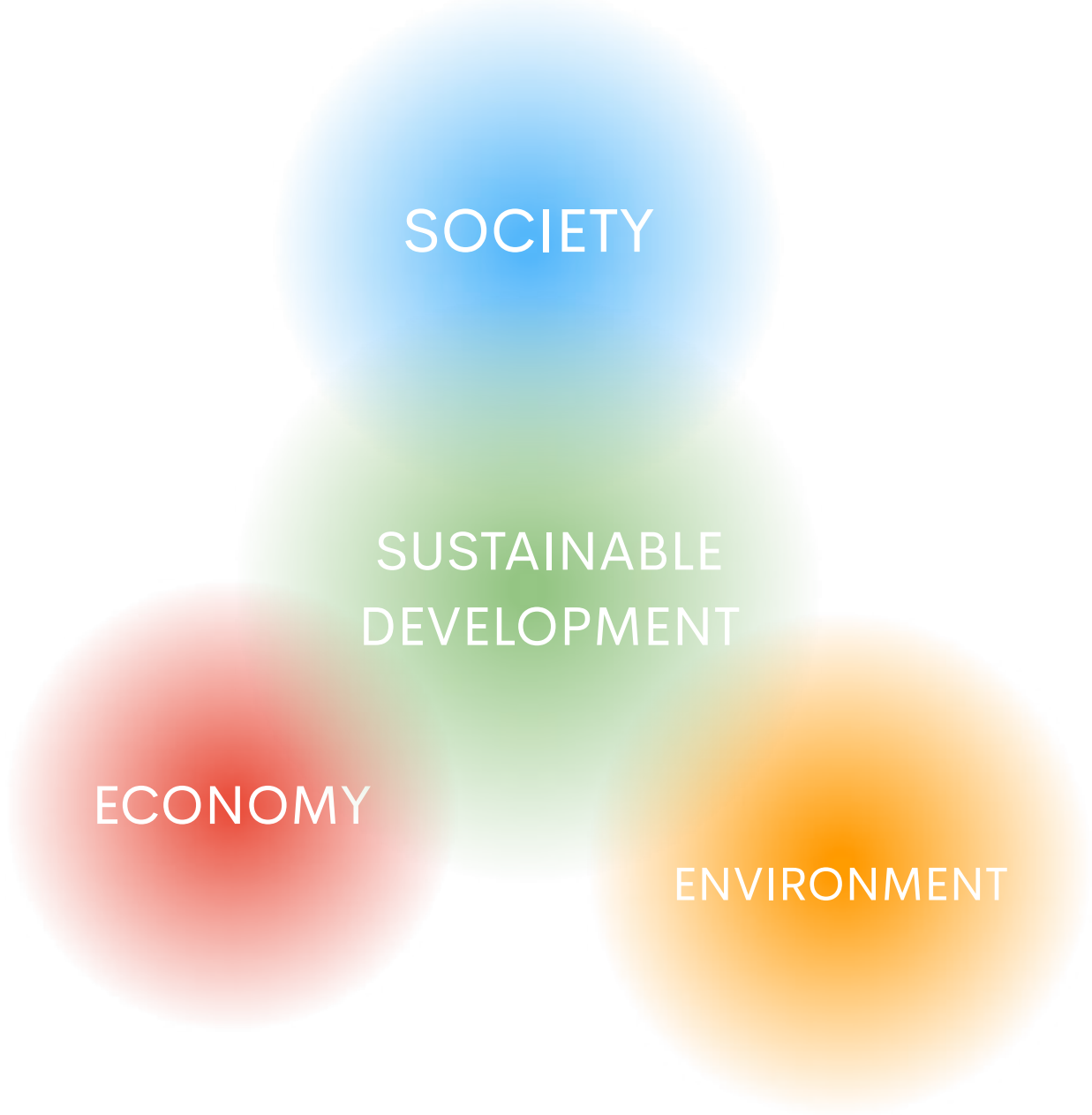
Strategy

KPI'S

CASE STUDY



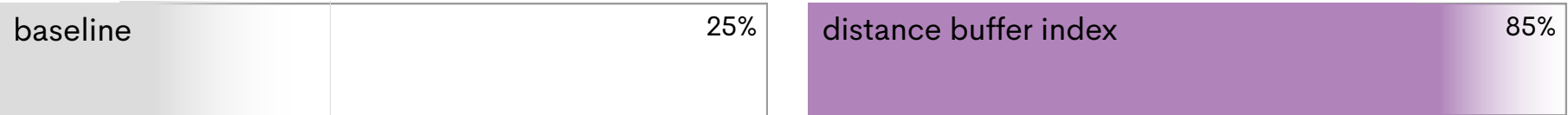
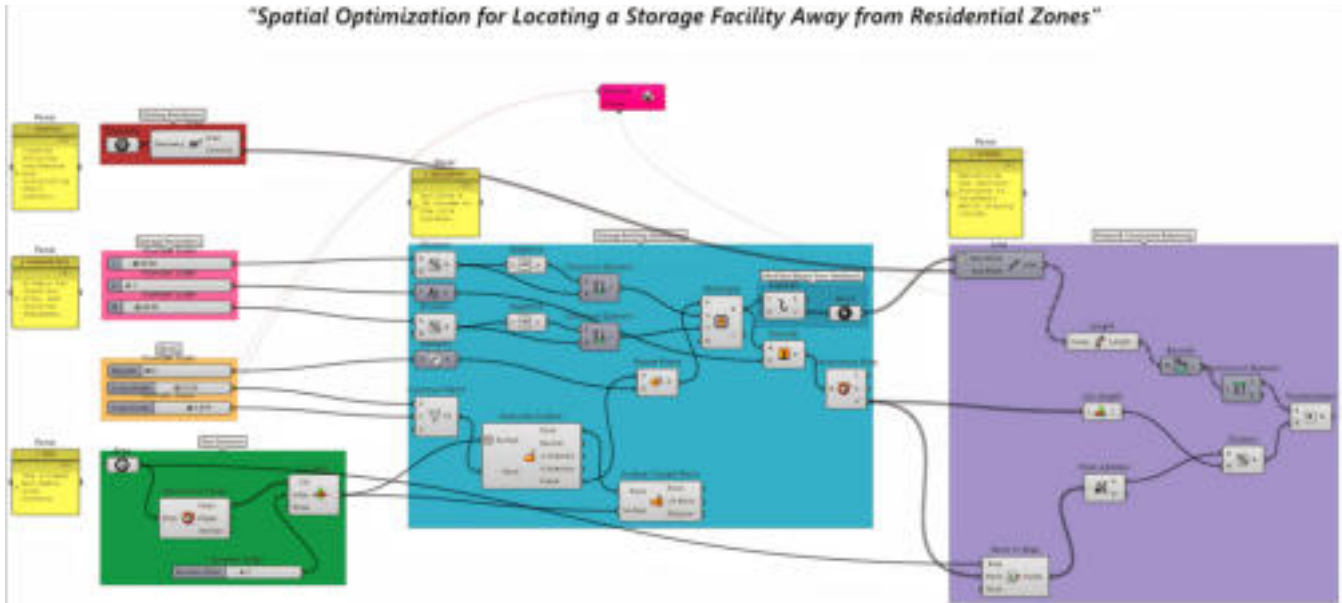
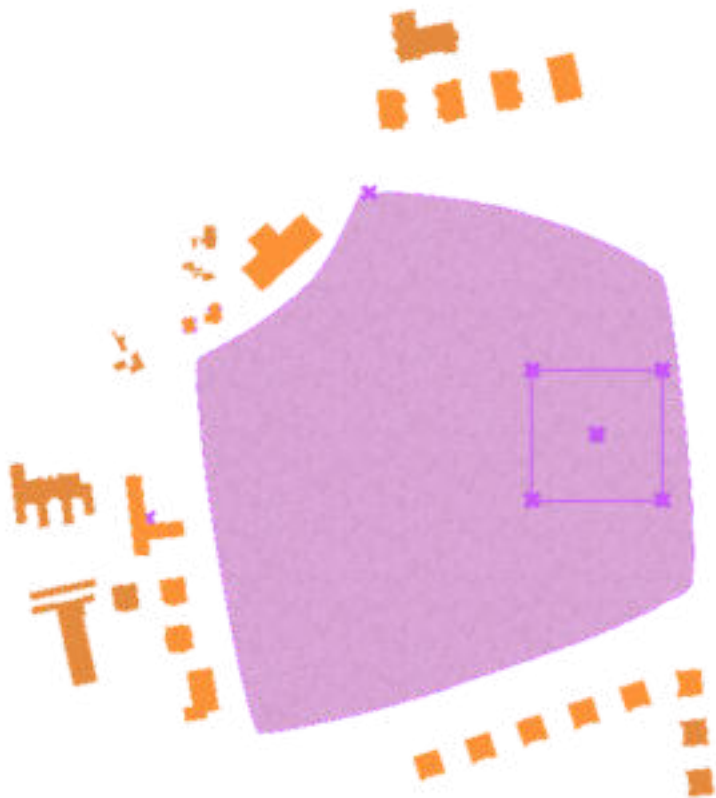
STRATEGY



To address the complex challenges of urban logistics, my design approach is anchored in the core principles of Sustainable Development. By purposefully integrating the three essential pillars Environment, Society, and Economy this strategy seeks to go beyond basic functionality. The goal is to create an architectural solution that resolves industrial inefficiencies while actively fostering ecological health, economic viability, and social well-being, ensuring the long-term resilience of the neighborhood.

URBAN

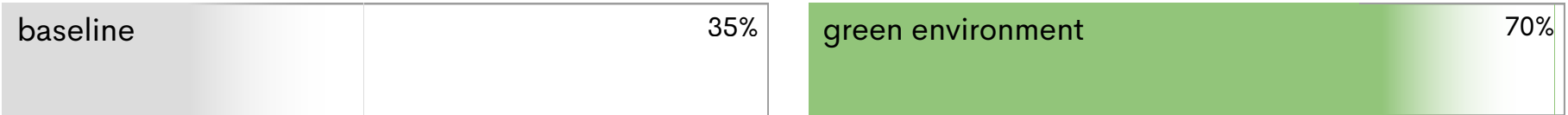
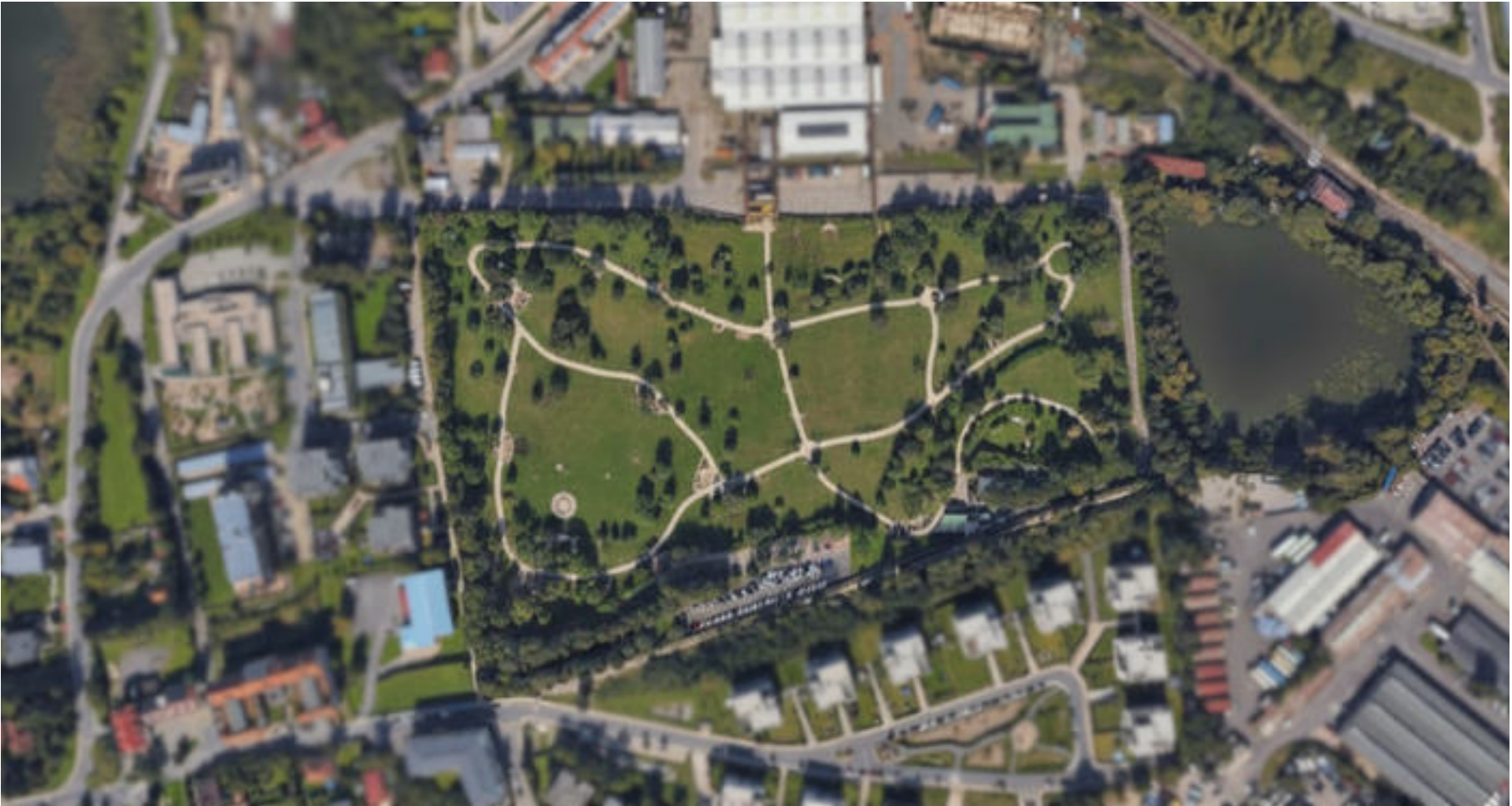
This urban optimization strategy calculates the maximum viable distance between a proposed logistics site and surrounding residential neighborhoods. By aiming to mitigate noise, traffic conflicts, and environmental disturbances, the model minimizes friction between industrial and domestic zones. Through testing multiple spatial iterations, the script generates a "distance buffer index," pinpointing locations that best isolate the facility from sensitive areas. Ultimately, this data-driven approach establishes an objective baseline for strategic site selection within complex urban fabrics.



GREEN

The "Green" strategy transforms a necessary spatial buffer into a thriving ecological corridor. Designed to maximize the distance between industrial operations and residential living, this zone is programmed as a landscaped park rather than empty terrain. Distinct pathways weave through the greenery, inviting the community to

engage with the natural environment while providing safe, walkable routes. This deliberate use of green infrastructure serves a dual purpose: it acts as a highly effective acoustic and visual shield against logistical disturbances, while simultaneously enriching the neighborhood with a vital, accessible natural amenity.



WATER

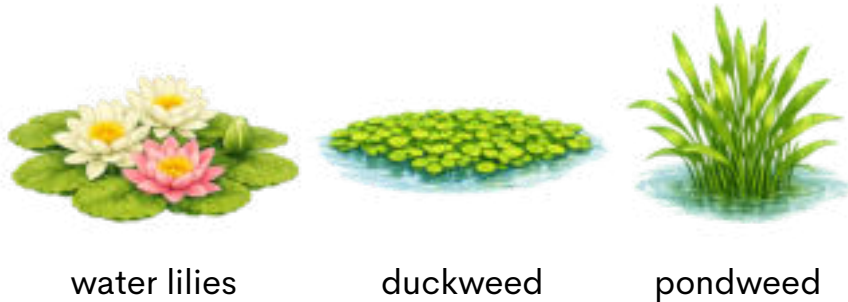
Water forms the vital foundation of a thriving lake ecosystem, supporting a complex web of life through carefully balanced plant diversity. Along the shallow margins, emergent vegetation such as reeds, cattails, and rushes play a crucial role by naturally filtering the water, stabilizing the shoreline against erosion, and providing essential nesting habitats for local wildlife. Just beyond the water's edge, the ecosystem transitions into the riparian zone, where moisture-loving plants like meadowsweet and marsh marigolds flourish. Together, these varied vegetation zones create a seamless and vibrant transition from water to land, significantly boosting overall lake biodiversity while offering crucial nectar sources and shelter for pollinators and amphibians alike.

emergent plants

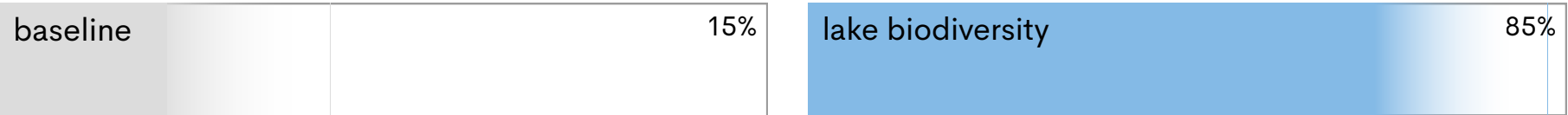
riparian edge plants



aquatic plants



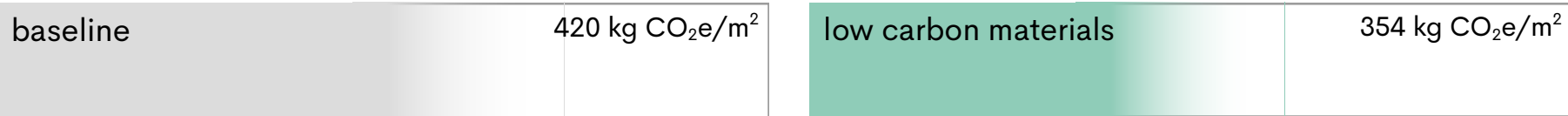
wet meadow plants



CARBON



The proposed design shifts away from traditional carbon-intensive logistics by prioritizing a Human-Centric Material Palette. By utilizing low-carbon concrete with 25% GGBS and structural reinforcement composed of 97% recycled content, the project significantly reduces the industrial 'weight' on the environment. This strategy achieves an impressive Carbon Class B rating, specifically mitigating the Social Cost of Carbon by over \$258,000. Beyond the numbers, these materials improve the local air quality and urban health of the Czech Republic site by minimizing toxic production phases (A1-A3). This data proves that 'Humanizing Logistics' is a tangible reality: by selecting materials that respect the Earth's limits, we create a built environment that is inherently safer and more sustainable for the people who inhabit it."



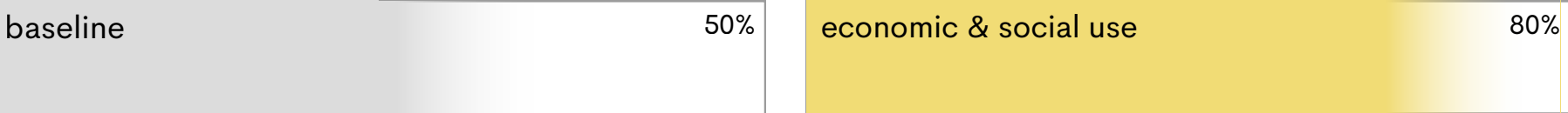
ENERGY

To truly "humanize" urban logistics, the architecture must take responsibility for its environmental footprint. This Energy Use Intensity (EUI) analysis breaks down the facility's operational demands across heating, cooling, lighting, and technology throughout the year. By understanding these seasonal energy peaks, the design strategically integrates extensive solar panel arrays across the sawtooth roof. This active renewable energy strategy offsets the building's consumption, ensuring the logistics hub operates as a sustainable, low-impact neighbor within the urban grid.



ECONOMY

Blurring the lines between heavy industry and civic life is essential to humanizing logistics. By situating local shops, markets, and small businesses along the facility's public-facing edge—directly interfacing with the adjacent park the architecture creates a dynamic economic transition zone. This mixed-use approach reclaims space typically lost to industrial sprawl, offering residents accessible retail and fresh food options while driving local commerce. Ultimately, this transforms a standard distribution center into a shared economic asset that benefits both the broader supply chain and the immediate community.



III.

SOLUTION

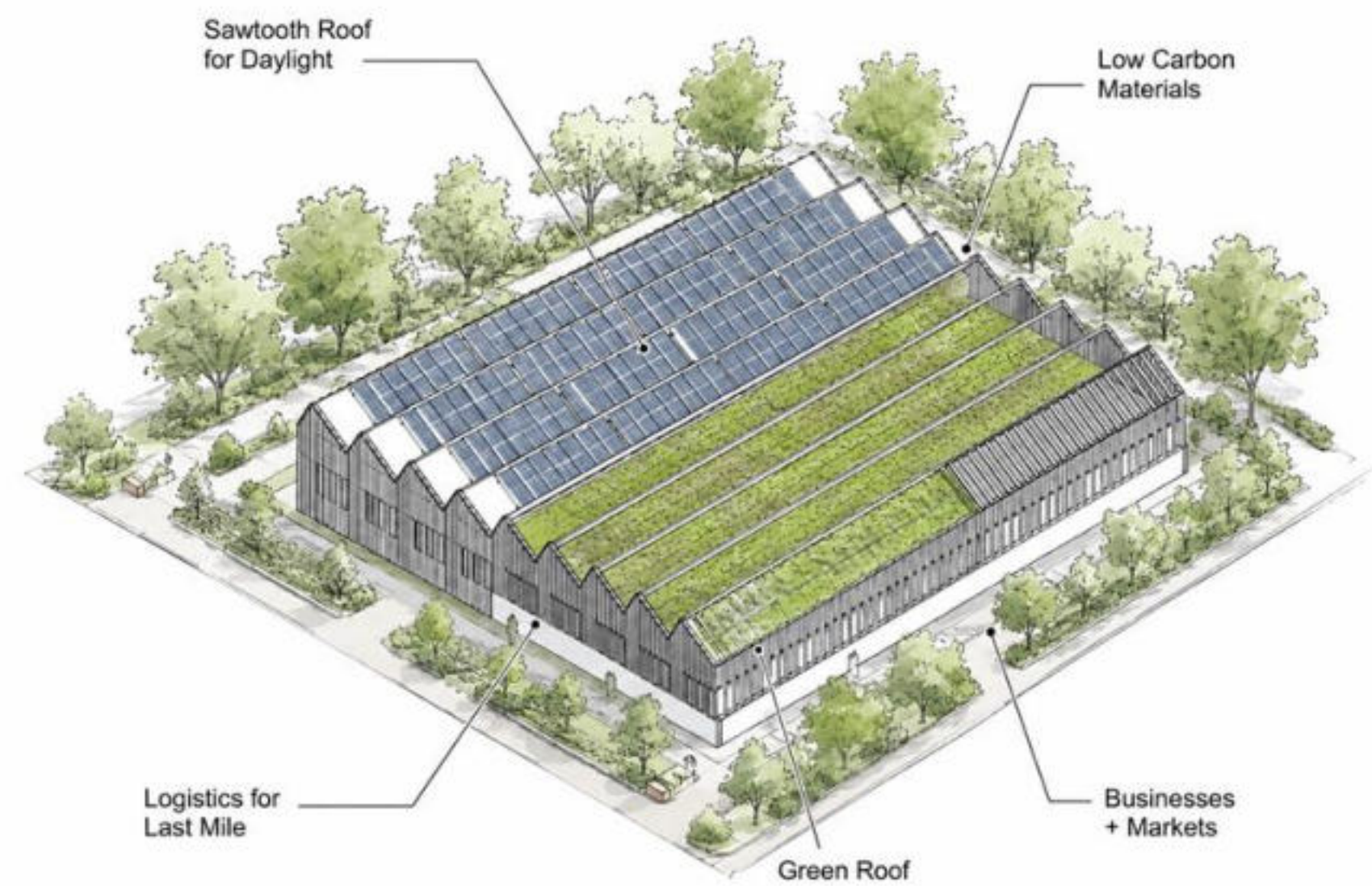
SKETCH

DIAGRAM

ELEVATION

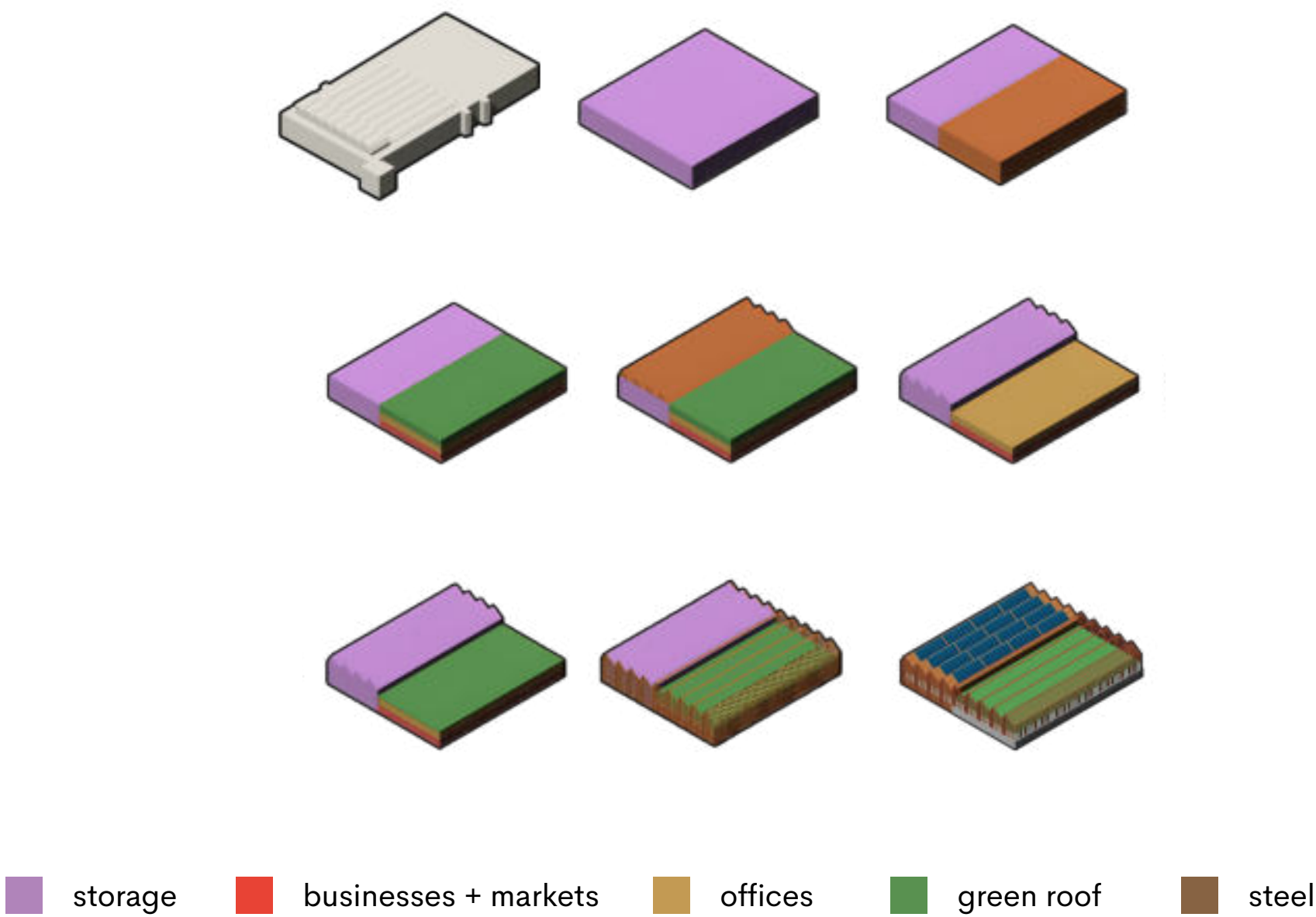
RENDER

SKETCH



The architectural design transforms the traditional 'logistics box' into a vibrant, community-oriented hub. By utilizing a sawtooth roof, the project maximizes solar energy and floods the interior with natural daylight, significantly improving the daily experience for workers. The integration of green roofs and native landscaping mitigates urban heat while providing a biophilic connection to nature. This approach proves that industrial infrastructure can be a social asset; by combining last-mile logistics with public markets and high-performance, low-carbon materials, we create a space that respects both the environment and the people who use it."

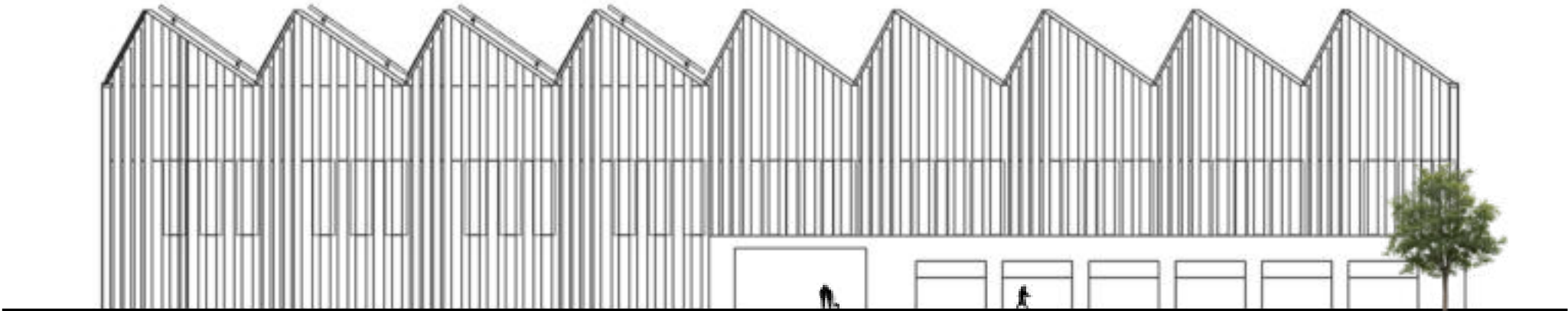
DIAGRAM



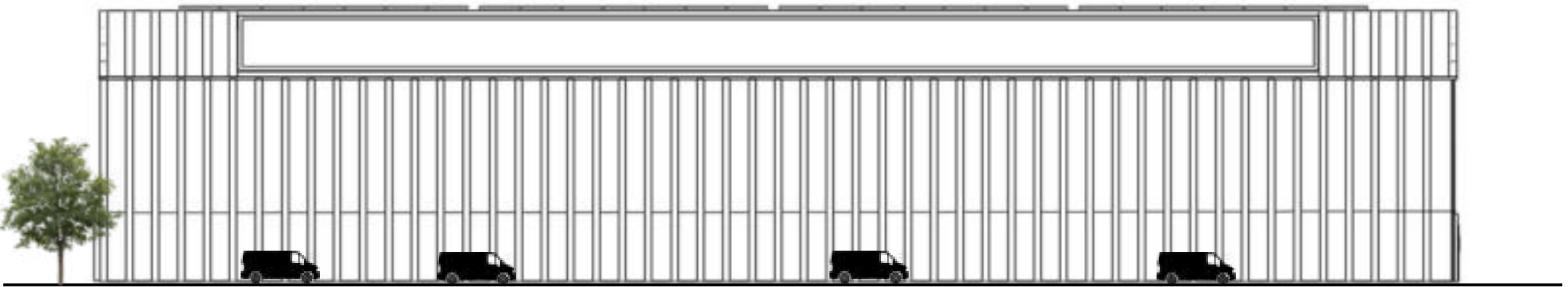
This diagram illustrates the sequential massing and programmatic distribution of the facility. Starting from the primary logistical footprint, the core volume is dedicated to essential storage. To break down the industrial scale and integrate with the community, human-centric programs offices, local businesses, and market spaces are introduced at the forefront. A generous green roof is then layered over these active zones, providing a vital ecological amenity. Finally, a distinct steel structural envelope ties the layered massing together, resulting in a cohesive, highly functional hybrid facility.

ELEVATION

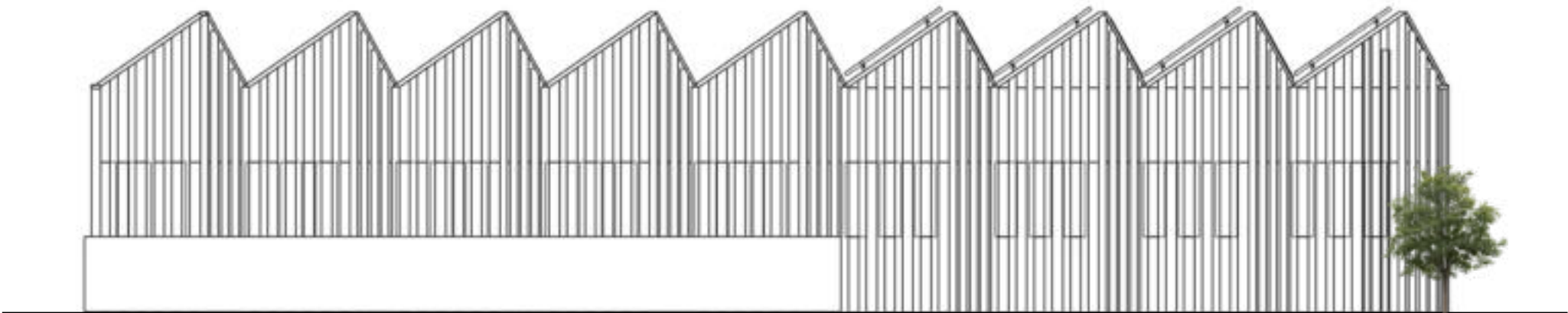
East Elevation



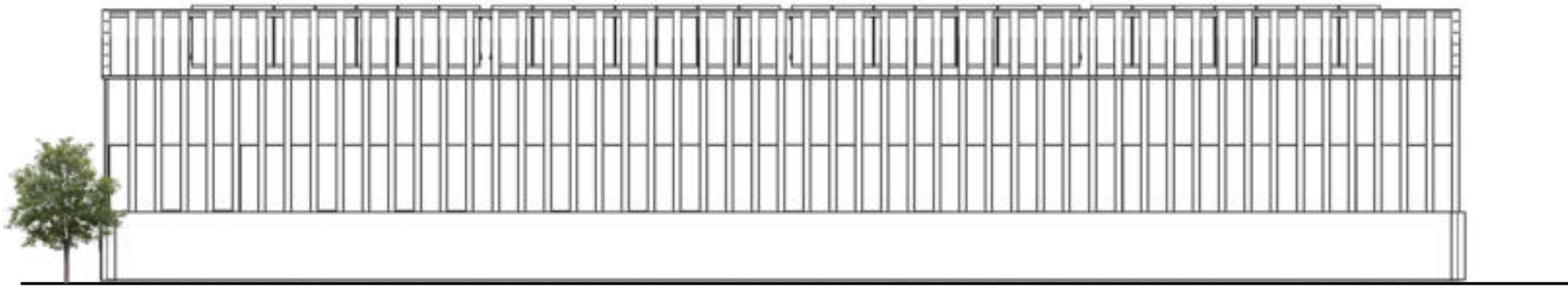
South Elevation



West Elevation



North Elevation



LAST-MILE LOGISTICS



BIODIVERSITY



GREEN

ECONOMIC AND SOCIAL USE



HUMANIZING LOGISTICS

This booklet demonstrates that logistics should no longer exist as isolated industrial systems disconnected from people and the city. Through data-driven design, environmental strategies, and human-centered planning, the project proves that logistics can become an integrated urban asset that supports ecology, economy, and community at the same time.

The research reveals how traditional logistics creates pollution, congestion, fragmentation, and environmental stress. Yet the data also proves that measurable interventions such as green corridors, renewable energy, biodiversity systems, low-carbon materials, and mixed-use public spaces can transform these facilities into healthier and more resilient environments.

As cities continue to grow and demand for last-mile logistics increases, rethinking logistics is no longer optional. It is necessary. The future of logistics must move beyond efficiency alone and instead prioritize people, sustainability, and coexistence with the urban fabric.

Humanizing logistics is not just an architectural idea it is a new vision for how industry, nature, and society can thrive together.

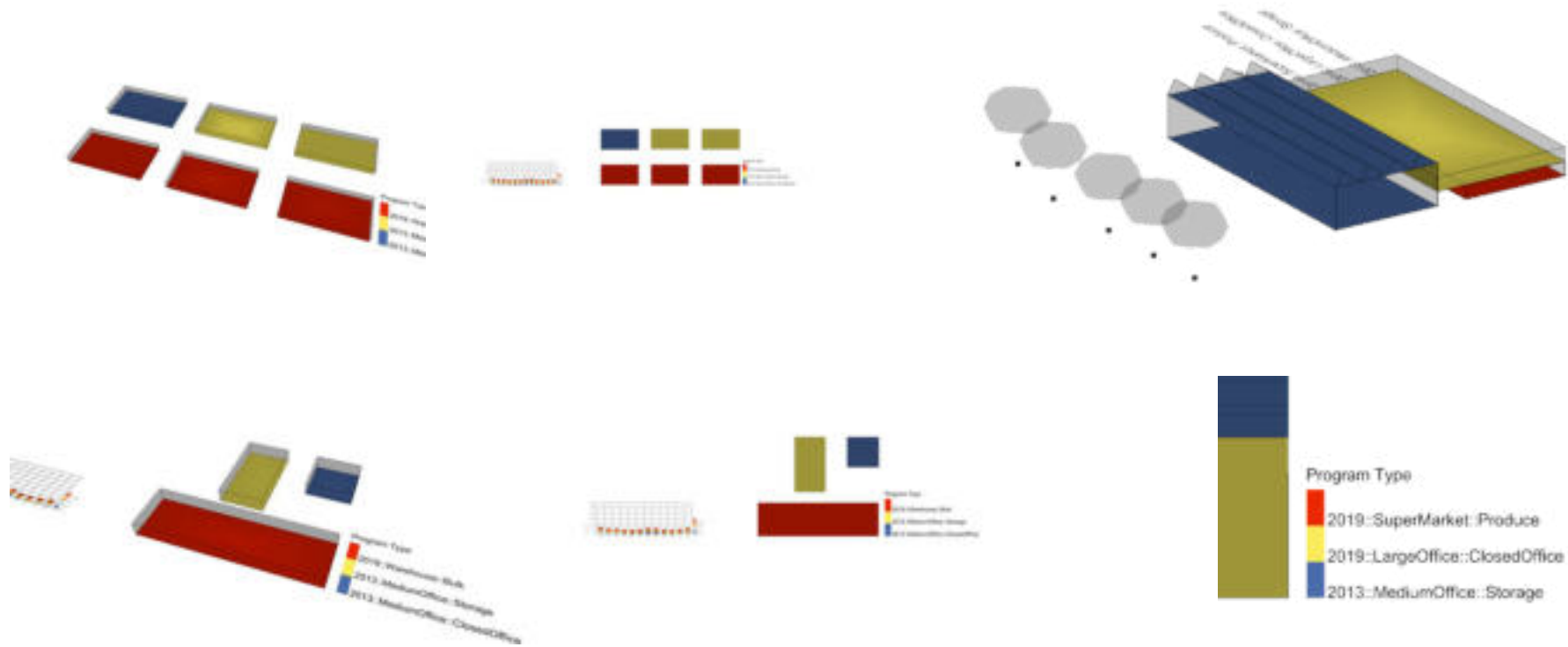


APPENDIX

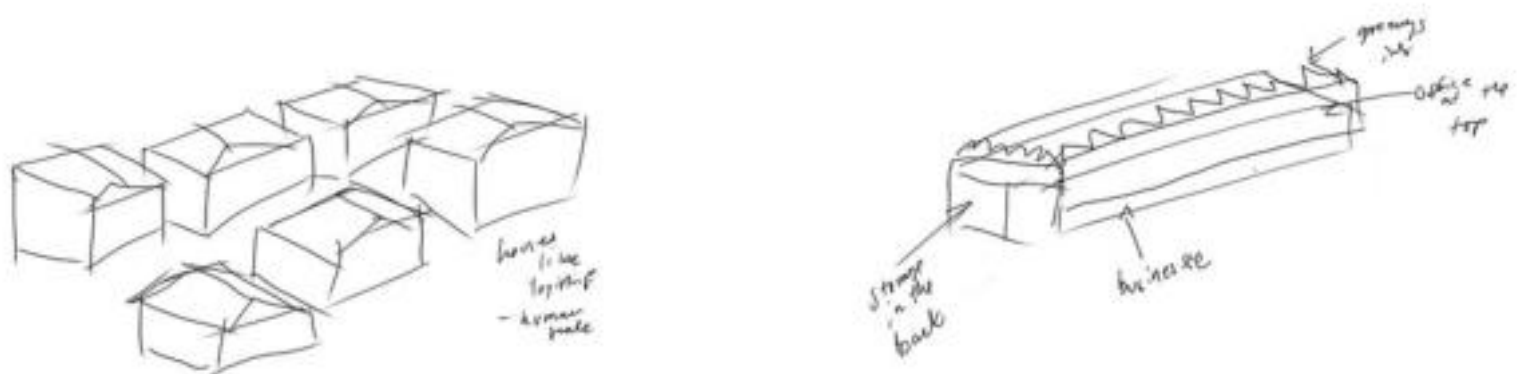
KPI'S



EUI



OPTIMIZATION SKETCH



LCA MODEL

